

EmoDrink: Embodied MR for Physiology-Informed Beverage Recommendation

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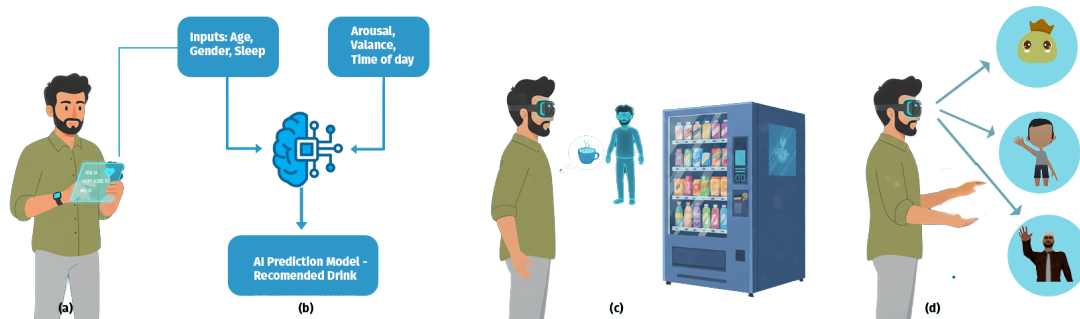


Figure 1: (a) User data collection via tablet input, self-reported sleep and mood, and physiological sensing. (b) A contextual model generates a drink suggestion from these inputs. (c) The user interacts with an embodied agent with a virtual vending machine. (d) Multiple visualizations present the recommendation

Abstract

Wearable devices increasingly capture physiological signals related to users' affective and cognitive states, yet these signals are commonly presented through numerical dashboards that are difficult to interpret and act upon in everyday contexts. We present *EmoDrink*, a Mixed Reality (MR) research framework for communicating physiology-informed wellbeing recommendations through embodied representations. EmoDrink integrates smartwatch-derived signals with brief self-reports and contextual cues to generate a single beverage recommendation, then holds this recommendation constant while varying how it is presented: an abstract visualization,

a generic embodied agent, and a personalized “future self” avatar. Through a walk-up-and-use MR experience, attendees can directly compare how embodiment shapes trust, comfort, and interpretability when engaging with identical wellbeing guidance.

CCS Concepts

• **Human-centered computing** → **Interaction design; Human computer interaction (HCI);**



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1 Introduction

Personalized recommendations increasingly mediate everyday food and beverage choices [5]. Recommender systems typically model preferences through explicit feedback or implicit behavioral cues, and may incorporate contextual information such as time and location [1, 2]. However, these approaches often overlook physiological signals that reflect momentary affective and cognitive states, despite the role emotions play in everyday choices (e.g., stress prompting calming options; fatigue prompting energizing, caffeinated options).

Wearable devices can continuously capture physiological and sleep-related signals, yet results are often surfaced as raw metrics or aggregate scores that are difficult to interpret and translate into actions [7, 8]. This creates a gap between sensing and decision-making in moment-to-moment contexts.

We present *EmoDrink*, an interactive MR framework that integrates physiological data (e.g., heart rate variability from a smartwatch), self-reported sleep and mood, and contextual cues to generate beverage suggestions. Critically, *EmoDrink* holds the underlying recommendation constant across three presentation strategies—an abstract visualization, a generic embodied agent, and a personalized avatar—so that embodiment becomes the primary interaction design variable. As a Demo experience, attendees can compare how these modalities shape interpretation of the same wellbeing advice within a virtual vending-machine scenario.

2 Background, Related Work, & Design Rationale

EmoDrink builds on personal informatics and embodied interaction research. Prior work shows that users often struggle to translate fragmented physiological metrics into actionable understanding, particularly in everyday, in-the-moment settings [6, 8]. Separately, research on embodied agents suggests that representation can shape engagement, trust, and perceived credibility compared to disembodied interfaces [3, 4]. *EmoDrink* connects these threads by treating embodiment not as a cosmetic layer, but as a key design variable for communicating identical physiology-informed recommendations.

Coarse Affective Model. *EmoDrink* uses a coarse arousal–valence model derived from physiological and contextual inputs (e.g., HRV, self-reported sleep quality, time-of-day patterns). This intentionally low-resolution inference avoids false precision, supports ambiguous affective states without forcing categorical labels, and frames outputs as reflective rather than diagnostic.

Vending Machine Scenario. A virtual vending machine provides a familiar, low-stakes decision context for presenting wellbeing recommendations while foregrounding differences in how recommendations are conveyed. Drinks are drawn from a curated set (e.g., calming, hydrating, energizing) to enable legible comparisons across modalities.

3 System Implementation

EmoDrink is implemented in Unity on a Meta Quest 3 with hand tracking enabled. The system comprises: (1) sensing and inference, (2) a recommendation backend, and (3) an embodied MR visualization layer (Figure 4).



(a) Virtual vending machine interface as seen through the head-worn display.



(b) Physical setup including head-worn display, wearable device, and tablet for contextual input.

Figure 2: *EmoDrink* interaction context. The system consists of a mixed reality vending machine interface and a physical setup integrating wearable physiological sensing and contextual user input.



(a) Abstract visualization. (b) Generic humanoid embodiment. (c) Personalized avatar embodiment.

Figure 3: Embodiment conditions used in *EmoDrink*. The same wellbeing recommendation is communicated through three representational modalities: an abstract visual form, a generic humanoid agent, and a personalized avatar.

Sensing, Inference, and Recommendation. Physiological signals are captured using a Garmin Venu 3S smartwatch and streamed via a companion application to a backend service, where signals are time-stamped and combined with brief self-reports (mood, sleep quality, energy) and contextual cues. A lightweight inference module estimates coarse affect (arousal/valence), and a recommendation engine produces a single beverage recommendation and structured

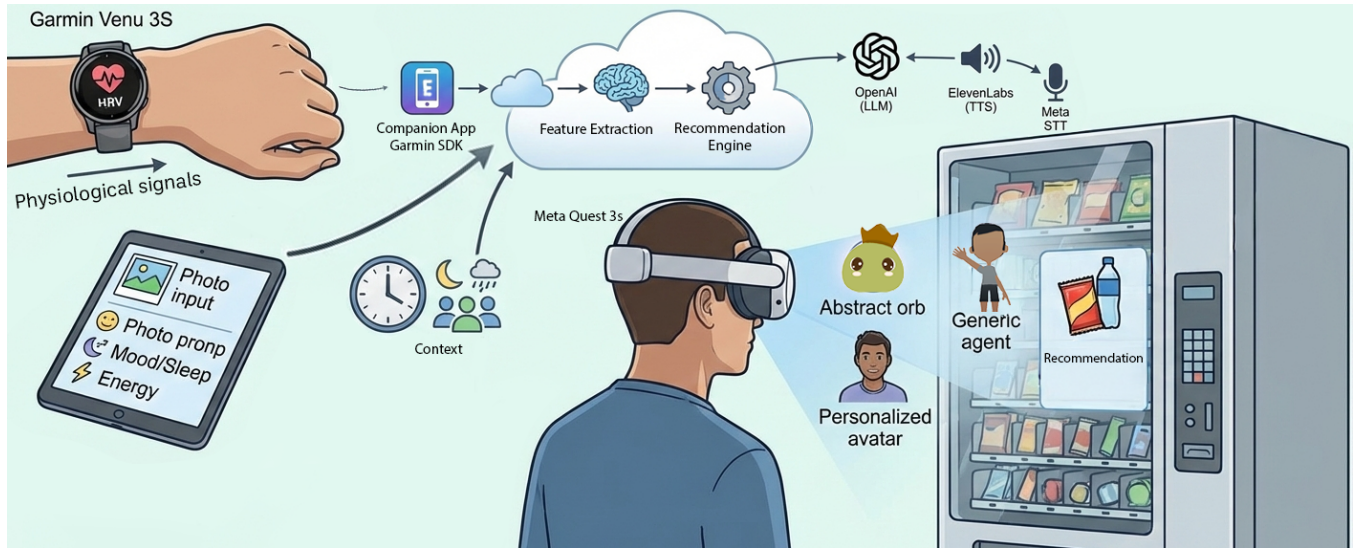


Figure 4: Physiological signals, self-reports, and contextual cues are fused by a recommendation engine and rendered in mixed reality through different embodiment strategies.

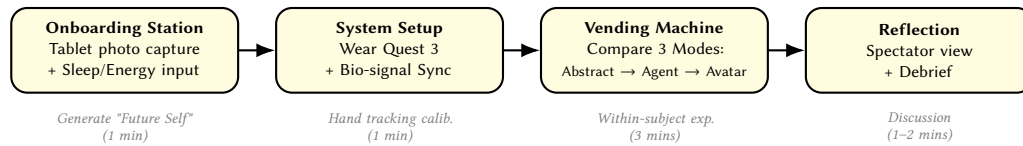


Figure 5: The EmoDrink Walk-up-and-use Demo Flow. Users create a personalized avatar, sync physiological data, and compare three embodiment strategies in MR.

rationale fields used for explanation.¹ The recommendation is held constant across all visualization modalities to isolate the effect of embodiment.

Dialogue and MR Rendering. The system generates natural-language explanations from the fixed recommendation and rationale fields using an LLM, and delivers them through speech (text-to-speech) with optional lightweight voice input (speech-to-text) for user acknowledgements and questions. The MR scene is rendered in passthrough using the Meta Passthrough API so the vending machine and embodied representations appear spatially situated in the user’s physical environment.

4 The Interactivity Experience

The demonstration offers attendees a 5–7 minute walk-up-and-use experience (Figure 5).

- **Onboarding (1–2 min):** Participants wear a smartwatch, provide brief self-reports (mood, sleep quality, energy), and take a tablet photo to generate a personalized avatar. These inputs produce a *single* wellbeing recommendation that remains fixed throughout the experience.
- **Guided Comparison (2–3 min):** Users enter MR and approach a virtual vending machine. They encounter the same

underlying recommendation through three embodiment modalities: an abstract visualization (data orb), a generic embodied agent, and a personalized avatar. Transitions are system-orchestrated to minimize interaction burden and foreground perceptual differences. The modality order is **randomized across participants**, while the recommendation remains unchanged.

- **Reflection & Exploration (1–2 min):** After experiencing all three modalities, users may optionally revisit any representation and explore explanation details (e.g., rationale or ingredient information) via voice or gesture.
- **Data Collection:** Participants complete a brief post-experience questionnaire assessing trust, clarity, comfort, and perceived actionability for each modality (Appendix A), along with preference ranking and optional short feedback. The system also logs basic interaction traces (e.g., modality order, time per modality, voice queries) to support analysis of within-subject comparisons.

5 Conclusion

EmoDrink explores how physiology-informed wellbeing recommendations can be communicated through embodied mixed reality interfaces, emphasizing *how* recommendations are presented rather

¹For details of the recommendation engine and feature design, refer to [9].

than *how accurately* affect is inferred. By holding the underlying recommendation constant and varying only embodiment, the system foregrounds representation as a core interaction design variable in affect-aware systems.

As a Demo experience, EmoDrink enables direct, within-subject comparison between an abstract visualization, a generic embodied agent, and a personalized avatar in a familiar, low-stakes vending-machine scenario. More broadly, EmoDrink functions as a modular research probe for studying embodiment, interpretability, and trust in future wearable and AR systems.

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A Post-Experience Questionnaire

Participants completed a brief post-experience questionnaire immediately after the EmoDrink interactivity. Unless otherwise noted, all items were rated on a 7-point Likert scale (1 = Strongly disagree, 7 = Strongly agree).

A.1 Per-Modality Ratings

The following items were completed separately for each of the three embodiment modalities: *Abstract Visualization*, *Generic Embodied Agent*, and *Personalized Avatar*.

- (1) **Trust:** “I trust the recommendation presented in this form.”
- (2) **Clarity:** “I understood why this recommendation was suggested.”
- (3) **Comfort:** “This representation felt comfortable rather than intrusive.”
- (4) **Actionability:** “This representation helped me decide what to do.”

A.2 Overall Comparison

- (5) **Preference Ranking:** “Rank the three representations by overall preference.”
(1 = Most preferred, 3 = Least preferred)
- (6) **Consistency Check:** “The recommendation remained the same across all representations.”
(Yes / No / Not sure)

A.3 Privacy and Reflection

- (7) **Privacy Concern:** “The personalized avatar felt privacy-invasive.”
- (8) **Open-Ended:** “Briefly explain why you preferred one representation over the others.”

In addition to self-report responses, the system automatically logged interaction metrics including time spent with each modality, order of modality exposure, number of voice queries, and modality revisits.